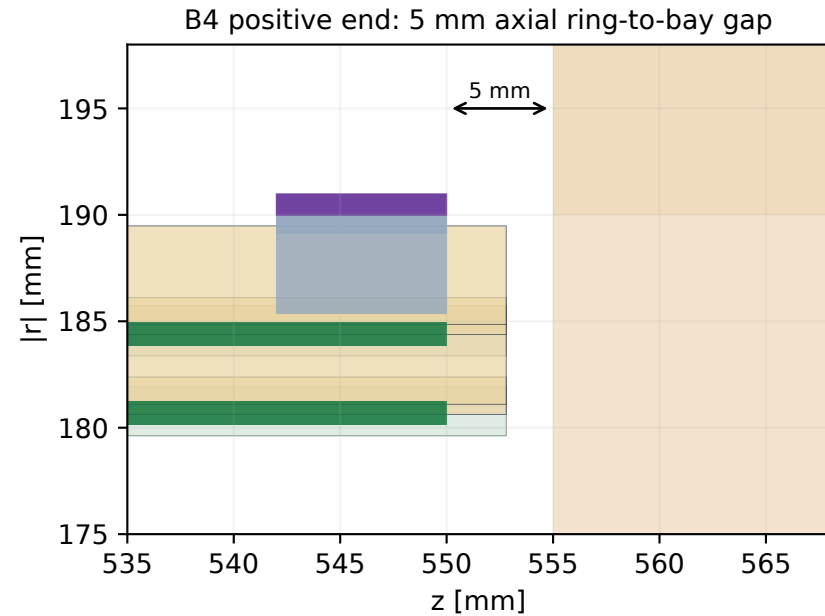
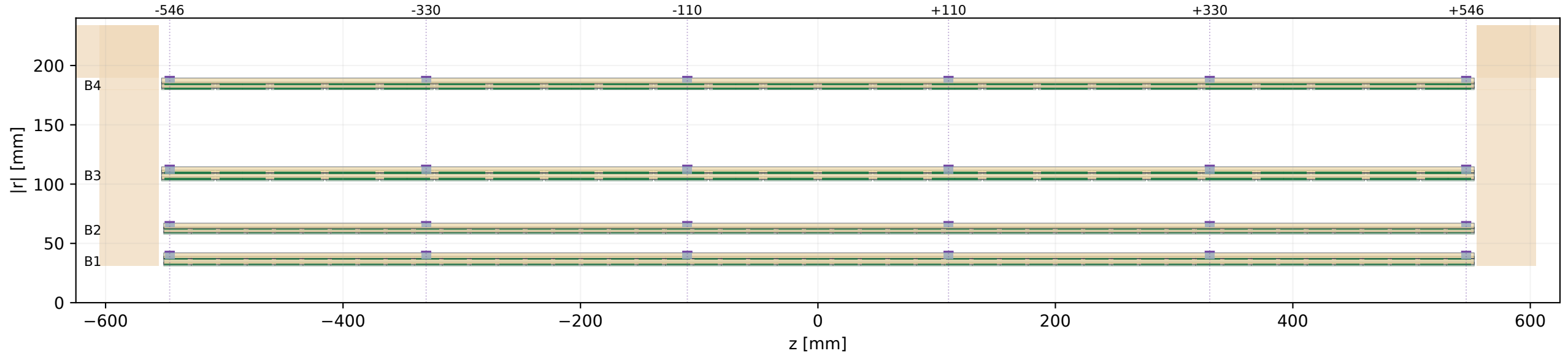


Candidate A — pixel barrel |r|-z assembly

Six ring stations per barrel, with end rings clear of the service bays

Projection over ϕ : both stagger levels shown; dotted vertical lines locate stations, not solid radial walls.



Mounting and constraint proposal

End rings at $z = \pm 546$ mm attach to global end supports (interface unresolved).
 Four internal rings hold barrel shape and share loads; maximum station pitch 220 mm.
 One axial locator at -546 mm; other feet slide/flex along z during cooldown.
 The global cage stiffness must be verified: short-span beam sag is not a cage result.

24 CFRP rings: 217.7 g; 492 solid foot envelopes: 81.1 g.
 Additional pins, inserts, joints, adhesive, flex and global mounts are not counted.
 B4 rings extend to $r = 190.983$ mm, only at the ring z bands.
 Their outer end is $|z| = 550$ mm; the reserved service bay begins at 555 mm.

- Active sensor plane (fixed)
- CFRP ring: 1 mm radial $\times$ 8 mm z
- Foam + CFRP stave (outward)
- Bearing-foot envelope (4 mm wide)
- Reserved service volumes